A Random Forest Approach to Voltage Deviation Management in Electric Vehicle Integrated Distribution Feeders

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***Abstract*—The widespread adoption of uncoordinated electric vehicle (EV) charging in homes, charging stations and wireless charging lanes introduces new power demands and operational dynamics in distribution grids that were not anticipated in distribution planning. Machine learning models offer a solution to the problems of managing the real-time dynamics of random charging patterns of multiple EVs in a distribution network. This paper seeks to demonstrate a hardware control prototype that performs this management autonomously at the grid edge, throttling EV charging current to arrest voltage sags in real time without reliance on a communications network. Specifically, it bridges the gap between software simulation and physical hardware constraints by deploying a Random Forest regressor directly onto a Texas Instruments C2000 embedded microcontroller. The Edge AI model utilizes only 65.0 KB of flash memory and executes a predictive inference in 176.88 microseconds. Upon predicting a severe voltage sag, the microcontroller adjusts a Pulse Width Modulation (PWM) output to smoothly throttle the EV charging rate rather than triggering a hard disconnect. Processor-in-the-Loop (PIL) validation demonstrates that this embedded intelligence correctly identifies impending voltage collapses and actively clamps the localized grid voltage at the 0.95 p.u. ANSI C84.1 Range A threshold, providing real-time deterministic grid stabilization.**

***Index Terms*—Electric Vehicles, Edge AI, Power Grid Resilience, Microcontrollers, Random Forest**

I. Introduction

The widespread transition towards sustainable energy sources and the shift away from fossil fuel usage has spurred the proliferation of electric vehicles (EVs) in many nations.

Previously, distribution grids were made to accommodate highly predictable cyclical residential and commercial loads. However, the adoption of electric vehicles has introduced an unprecedented variable: random, high-intensity, and closely timed power demands [1]. Uncoordinated EV charging represents a threat to power quality, as EV chargers function as constant power loads that draw large amounts of current over extended durations [2], [3].

These effects are compounded by the constant-power characteristic of modern electronic loads and chargers, which respond to a voltage sag by drawing more current — reinforcing the sag rather than damping it. If left unmitigated, these voltage sags lead to thermal stress, flicker and possible brownouts.

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To maintain power quality and prevent equipment damage, utility operators require that distribution networks strictly adhere to the ANSI C84.1 Range A lower voltage tolerance boundary of 0.95 per-unit (p.u.) [4], [5].

Mitigating these sags is fundamentally a problem of both speed and locality: a coordinated surge in demand, such as coincident EV charging, can drive a node toward collapse faster than a centralized controller can observe, decide, and then issue a response. Effective intervention therefore must sense the local grid state and actuate a correction autonomously. Conventional stability-management approaches instead concentrate their intelligence centrally, depending on Wide Area Measurement Systems (WAMS) and cloud-based computation whose communication latency times are poorly matched to real-time distribution control [6]. This gap between the capabilities of cloud-based infrastructure and the demands of an EV-dense grid motivates relocating predictive intelligence from the cloud to the grid edge itself.

Random Forest methods in particular have shown strong performance on grid problems, including static voltage stability assessment [7] and voltage stability analysis spanning transmission and distribution networks [8]. These efforts, however, were developed for centralized deployment, drawing on wide-area measurements and off-device computation rather than autonomous operation at a single distribution node. Their reliance on network-wide data and back-end processing leaves the problem of fast, self-contained voltage mitigation at the grid edge largely unaddressed.

This paper presents a hardware-based methodology: the deployment of an Edge AI framework operating on a Texas Instruments C2000 microcontroller. By embedding a Random Forest algorithm [9] onto the real-time controller, the system factors in the fixed localized line impedance while actively monitoring baseline voltage and dynamic EV current flow to mitigate voltage sags in real-time.

II. Proposed Methodology

A. Dataset Generation and Physics Modeling

Since grid voltage drops are consequences of specific grid states, the training dataset was constructed using standardized network test cases distributed with the MATPOWER framework [10]. To prevent topological bias, the machine learning model was not constrained to a single test feeder; rather, a diverse set of peer-reviewed network geometries — including the IEEE 14-bus transmission system and the 33-bus radial distribution system — was simulated to diversify the impedance and voltage regimes represented in the training data. This multi-topology approach provided a pre-validated foundation of the physical rules for the networks, yielding realistic baseline line impedance and grid voltage parameters across multiple configurations. To simulate the impact of EVs, a Python-based batch simulation framework was engineered to inject random EV load profiles (P_{EV}, Q_{EV}) into the nodes of these established cases. By iterating this process across thousands of randomized permutations, the simulation generated a comprehensive feature space. The target variable for the machine learning model was the predicted grid voltage movement under specific loading conditions, specifically for identifying the points where the combination of the MATPOWER defined impedance and the EV power draw forced the voltage below a typical regulatory 0.95 p.u. threshold.

B. Random Forest Algorithm

The Random Forest algorithm was optimized for the computational limitations of the target embedded hardware. The model was configured with 30 decision trees, with a maximum depth of 5 node splits per tree. By enforcing these limits, the model achieved statistical accuracy while keeping the memory at an acceptable size for the C2000 microcontroller's memory allocation. The maximum depth constraint ensures that the worst-case inference path requires no more than 5 consecutive conditional evaluations, ensuring adequate computational speed.

C. Offline Training and Validation Metrics

The Random Forest model was implemented, trained, and validated in Python using the scikit-learn library [11]. The simulated dataset was partitioned into separate training and held-out test subsets, and the predictive performance on the held-out data was quantified using a coefficient of determination (R^2), mean absolute error, and mean squared error, as presented in Table I.

TABLE I
Offline Model Validation Metrics

Evaluation Metric	Measured Value	Physical Unit
R-Squared Score	0.9769	Dimensionless
Mean Absolute Error (MAE)	0.0017	p.u.
Mean Squared Error (MSE)	7×10^{-6}	p.u.^{2}

An R^2 value of 0.9769 indicates that the Random Forest model accurately captures 97.69% of the variance in the grid's voltage drop behavior. The low mean absolute error and mean squared error also confirm that the predictive error margin is minimal, preventing premature mitigation triggers while operating near the 0.95 p.u. threshold.

III. Hardware Implementation

A. Physical Constraints of the C2000 Microcontroller

The Texas Instruments C2000 microcontroller architecture contains a 32-bit floating point CPU that was optimized for low latency and extremely fast real-time processing [12]. However, it lacks a traditional operating system and dynamic memory management. The machine learning model had to be translated into raw code that operates efficiently within the very limited flash memory and SRAM without interfering with primary tasks such as pulse width modulation generation or analog-to-digital conversions.

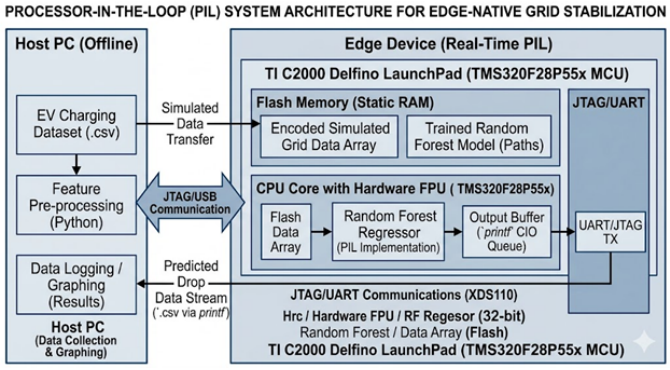
B. The Porting Process and Processor-in-the-Loop (PIL)

The trained Python model was mapped into typed format by using EloquentML, a Python library. The structure of the decision trees was translated into a standalone C++ header file, converting the machine learning model into a long set of if-else controls. The static structure allows the C2000 compiler (Code Composer Studio) to heavily optimize the branch predictions at a very low level.

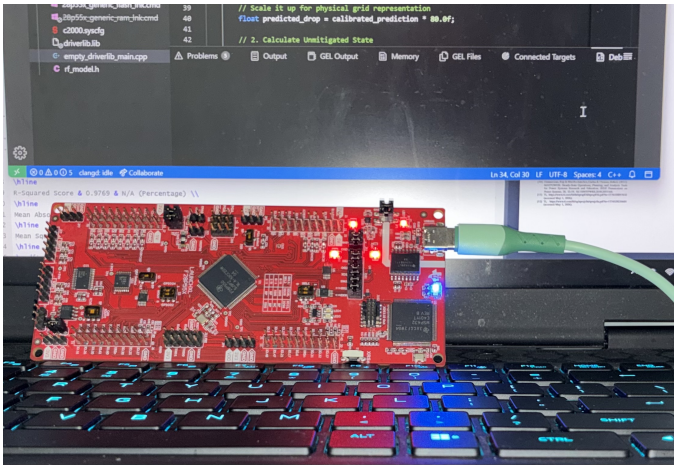
To safely validate the system, a PIL testing framework was developed. The physical plant (distribution grid mechanics and EV loading profiles) is mathematically simulated on a host PC. Rather than relying on physical analog voltages to be injected into the C2000's pins, the simulated signals are transmitted digitally to the C2000 via a communication link (USB). The physical C2000 processor receives this digital signal directly in its SRAM, executes the Random Forest prediction on its chip, and outputs a digital control signal back to the PC. This process isolates the computations of the microcontroller, closing the PIL feedback loop and proving viability without introducing external electrical sensor noise. The complete architecture and its physical realization are shown in Fig. 1.

C. Predictive Control Theory and Actuation

A notable advantage of the proposed embedded predictive controller over traditional reactive controllers is the elimination of sharp oscillations, where charging is abruptly stopped then completely restarted. A standard reactive logic controller that is programmed to disconnect an EV load only after crossing the 0.95 p.u. safety threshold starts a violent cycle: the load is dropped, the grid voltage is instantly spiked back up to 1.00 p.u., then the EV reconnected and the draw resumes, and the voltage violently crashes again. This happens repeatedly, and the rapid oscillation degrades power quality via voltage flicker, while the unmitigated load risks exceeding transformer ratings. Conversely, the proposed controller operates as a predictive closed-loop system. By continuously monitoring the localized impedance and current draw, the algorithm predicts the magnitude and direction of the voltage drop before it



(a) PIL system architecture for edge-native grid stabilization.



(b) The TI C2000 LaunchPad interfaced with the host PC over a USB serial link.

Fig. 1. Processor-in-the-Loop (PIL) validation environment.

occurs. Instead of triggering a complete disconnect, the microcontroller dynamically changes the duty cycle of its PWM output. This PWM signal acts as a throttle, smoothly choking the charging current rather than completely disconnecting the draw. The controller continually adjusts this throttle to maximize power delivery, ensuring that the EV can draw the maximum current physically possible without dragging the localized grid below the ANSI C84.1 Range A 0.95 p.u. threshold.

IV. Results & Discussion

A. Edge Hardware Feasibility

The memory footprint and inference latency were profiled directly on the C2000 LaunchPad development kit [13].

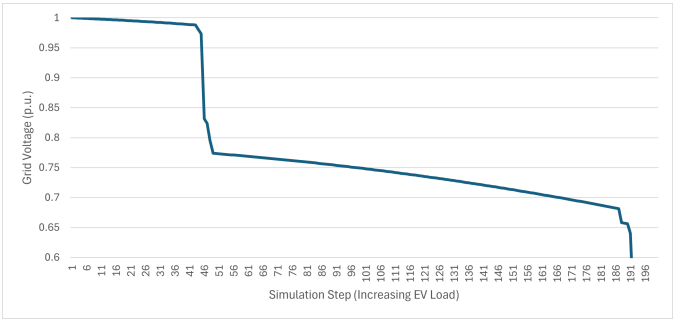
TABLE II
Texas Instruments C2000 Resource Usage

Hardware Resource	Available Capacity	Allocated by Model	Utilization (%)
Flash (ROM)	262 KB	65.0 KB	24.8%
SRAM	94 KB	14.7 KB	15.6%

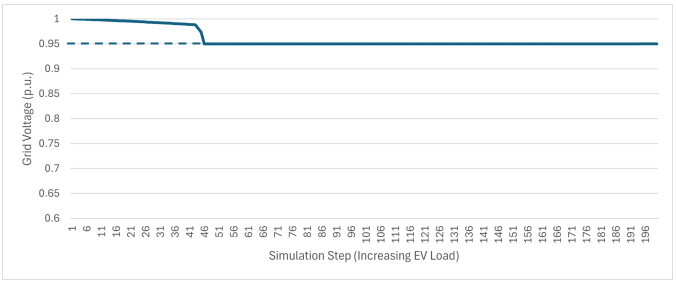
As summarized in Table II, the entire Random Forest model occupies only 65.0 KB of flash memory and 14.7 KB of SRAM, confirming that the model fits on the chip without exhausting the resources required for other processes, such as pulse width modulation and analog-to-digital conversion.

B. Voltage Mitigation Performance

The actual electrical effectiveness of the framework was tested via PIL simulation. A large, simulated fleet of EVs was suddenly connected to the network, sharply increasing current demand. Two identical grid simulations were run: a control group, without the mitigation program active, and an experimental simulation with the algorithm active.



(a) Unmitigated control group.



(b) Mitigation enabled.

Fig. 2. Comparison of localized grid voltage response to sudden EV load connection.

As shown in Fig. 2, both networks experience normal power fluctuations near 1.00 p.u. until the heavy EV load is connected. This instant spike in current causes the voltage to rapidly decrease. In the unmitigated control group, the voltage undergoes a complete collapse due to uncompensated losses, plummeting far below the 0.95 p.u. threshold.

Conversely, the mitigation-enabled hardware demonstrates the validity of the algorithm. When the system undergoes physical disturbance, the C2000 processes the incoming signals in 176.88 microseconds. Recognizing the signature of an impending collapse, the embedded controller outputs a mitigation signal. This quick intervention immediately stops the downwards trajectory of the voltage by throttling current draw, successfully stopping the voltage at the 0.95 p.u. threshold. The mitigated voltage does not flatline; rather, it forms a ripple along the threshold line. These tiny oscillations confirm that the embedded controller is actively functioning as a closed-loop controller, continuously recalculating the impending drop

and adjusting the PWM duty cycle to allow the maximum safe current draw while keeping the localized grid voltage above the 0.95 p.u. threshold.

V. Conclusion

The transition towards highly decentralized power grids requires control frameworks that operate autonomously under physical hardware constraints. This paper demonstrates the deployment of an accurate machine learning model onto a real-time embedded microcontroller to mitigate EV-induced voltage drops. By translating an optimized Random Forest model into static C++ code, the mathematical accuracy ($R^2 = 0.9769$) was preserved. Hardware validation proves that this model can execute natively on a Texas Instruments C2000 microcontroller, consuming just 65.0 KB of flash memory with a latency of 176.88 microseconds. By residing directly on the chip, the Edge AI bypasses the latencies of network communication, clamping localized voltages at the 0.95 p.u. threshold and preventing EV-induced voltage collapse. This hardware implementation establishes that Edge AI is a valid solution for advancing autonomous smart grids.

PIL validation, however, relies on ideal digital data transfer over a serial link, whereas physical power electronics must operate amid analog signal noise, ADC quantization error, and wire propagation delays. Future work will therefore advance from the PIL framework to a full real-time Hardware-in-the-Loop (HIL) testbed, in which the simulated impedance and current profiles are routed through a physical grid simulator that drives real analog voltages into the microcontroller's ADC pins. The C2000 would then perform the conversion, execute the Random Forest inference, and return PWM actuation to the simulator, providing definitive validation of the system's robustness under non-ideal physical conditions.

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